

Deteccção de Anomalias utilizando Redes Sem Pesos no dataset TON IoT

Grupo 2

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Desafio e Datasets

- **Artigo: TON_IoT Telemetry Dataset: A New Generation Dataset of IoT and IIoT for Data-Driven Intrusion Detection Systems**
- Detecção de anomalias em Sistemas de Detecção de Intrusão (IDS) voltados para ambientes IoT e IIoT.
- Dados de sensores disponíveis:
 - **Fridge** (1 contínua, 1 binária)
 - **Garage Door** (2 binárias)
 - **GPS Tracker** (2 contínuas)
 - **Motion Light** (2 binárias)
 - **Modbus** (4 contínuas)
 - **Thermostat** (1 contínua, 1 binária)
 - **Weather** (3 contínuas)



Metodologia

- **Classificação binária de anomalias**
- Modelos sem peso avaliados
 - **WiSARD, ClusWiSaRD, Bloom WiSARD, BTHOWeN, ULEEN**
- Tratamento de Dados
 - Date & Time -> Removidos para evitar vazamento
 - Label (0=Normal, 1=Ataque) -> Usados para classificação binária
 - Type (normal, backdoor, ddos, injection, xss, ...) -> Não utilizados
- Otimização de hiperparâmetros
- Divisão do dataset: **80%** Treinamento | **20%** Teste
- Comparação dos resultados dos modelos
 - **Métricas: Acurácia, Precisão, Recall, F1-score**



Estratégias de Binarização

- Variáveis Categóricas
 - Binarização Direta (0 e 1)
 - One-Hot Encoding
- Variáveis Numéricas
 - Termômetro Gaussiana
 - Termômetro Logarítmico
 - Termômetro Circular



WiSARD

- **Binarização**
 - Variável categórica de dois valores -> **Binarização Direta (0 e 1)**
 - Variável numérica (float ou int) -> **Termômetro Gaussiano**
- **Hiperparâmetros avaliados**
 - N° bits do termômetro: 4, 8, 16
 - N° bits da tupla: Todos os divisores do total de bits até 24
 - Ignorar tuplas zeradas: True, False
 - Bleaching



WiSARD

Dataset	Accuracy	Precision	Recall	F1-score	Train time (s)	Test time (s)	Tuple	Ignore Zero	Bleach	Bits/term
Fridge	0.7671	0.3868	1.0000	0.5578	0.18	1.04	2	True	16246	4
Garage_door	0.4979	0.1283	0.5019	0.2044	0.08	0.74	1	True	1837	8
Gps_tracker	0.8166	0.3082	0.2688	0.2872	0.80	2.34	1	False	13912	16
Modbus	0.5110	0.2269	0.4913	0.3104	0.10	0.47	16	True	72	4
Motion_light	0.5001	0.1408	0.4971	0.2194	0.07	0.58	1	True	3871	16
Thermostat	0.5016	0.1274	0.4985	0.2029	0.11	0.76	1	True	3197	4
Weather	0.5937	0.1941	0.6086	0.2943	0.17	1.13	24	False	99	8
Global	0.6393	0.3499	0.3822	0.3653	0.47	0.92	14	True	23948	4



ClusWisard

Pré-processamento

- **Variáveis contínuas -> Linear (Termômetro)**
 - Temperaturas (Fridge, Thermostat, Weather), Pressão/Umidade (Weather), Sensores (Modbus).
- **Binarização direta**
 - Condições e Estados (Fridge, Garage Door, Motion Light, Thermostat), Sinal de telefone (Garage Door).
- **Coordenadas -> Circular**
 - Latitude e Longitude (GPS Tracker).

Grid Search para Parâmetros

- resolução termômetro: 64, 128, 256
- address_size: 4, 6, 8, 12, 16
- min_score: 0.05, 0.1, 1.5
- threshold: 5, 10
- discriminator_list: 5, 10, 20



ClusWisard - Resultados

Dataset	Accuracy	Precision	F1-score	Recall	Train time (s)	Test time (s)	Address Size	Min Score	Threshold	Disc. Limit
IoT Fridge	0.50	0.15	0.22	0.49	4.89	30.57	4	0.05	10	5
IoT Garage Door	0.19	0.14	0.24	1.00	0.12	0.12	4	0.05	5	5
IoT GPS Tracker	0.54	0.22	0.36	0.94	0.99	1.64	12	0.15	10	5
IoT Motion Light	0.14	0.14	0.25	1.00	0.08	0.08	4	0.05	5	5
IoT Modbus	0.54	0.26	0.37	0.59	24.32	26.63	16	0.05	10	20
IoT Thermostat	0.62	0.13	0.19	0.35	2.14	15.60	4	0.05	5	5
IoT Weather	0.86	0.50	0.24	0.16	5.11	71.29	0	0.00	0	0
TON IoT Geral	0.70	0.43	0.34	0.28	7.30	817.10	4	0.05	10	5



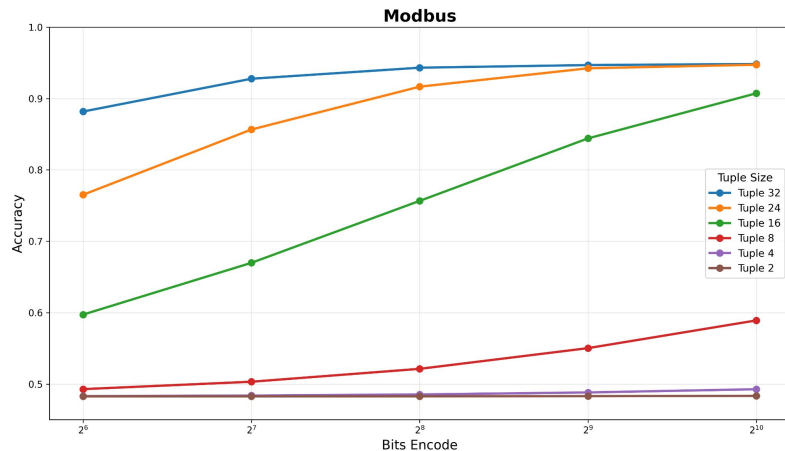
BloomWisard

Pré-processamento

- `fillna(0)` para valores ausentes
- Binarização usando **Thermometer Encoding**

Configurações Avaliadas

- `bits_encoding`: 64, 128, 256, 512 e 1024
- `tuple_bit`: 2, 4, 8, 16, 24 e 32
- Taxa de erro Bloom Filter: 0.01





Resultados do BloomWisard

Dataset	Accuracy	Precision	Recall	F1-score	Train time (s)	Test time (s)	Bits/encoding	Tuple bit
fridge	0.5085	0.6292	0.5186	0.5684	8.22	3.64	1024	32
garage_door	0.3803	1.0000	0.0022	0.0045	3.05	1.30	2	2
gps_tracker	0.7316	0.9974	0.5651	0.7215	0.97	0.30	256	32
modbus	0.9847	0.9924	0.9780	0.9851	3.90	1.20	1024	32
motion_light	0.3815	0.6667	0.0053	0.0105	0.97	0.30	2	2
thermostat	0.4577	0.2095	0.4577	0.2874	0.24	0.07	256	24
weather	0.8148	0.9916	0.7063	0.8250	0.77	0.23	256	32
combinado	0.8320	0.8697	0.4489	0.5922	33.59	11.24	256	32



BTHOWeN

- **Binarização**
 - Variável categórica de dois valores -> **Binarização Direta (0 e 1)**
 - Variável numérica (float ou int) -> **Termômetro Gaussiano**
- **Hiperparâmetros avaliados**
 - N° bits do termômetro: 4, 8, 16
 - N° bits de entrada por filtro de bloom: 4, 8, 16, 24, 32
 - N° de posições de memória por filtro de bloom: 128, 256, 512, 1024, 2048, 4096
 - N° de funções hash: 2, 3
 - Bleaching



BTHOWeN - Resultados

Dataset	Accuracy	Precision	Recall	F1-score	Train time (s)	Test time (s)	Bits/term	Bits/filtro	Entries/filtro	Bleach	Hashes
Fridge	0.7672	0.3870	1.0000	0.5580	1.47	3.01	8	8	128	11398	2
Garage_door	0.5020	0.1297	0.5033	0.2062	0.97	3.33	4	2	128	22113	3
Gps_tracker	0.7116	0.2074	0.3896	0.2707	1.40	4.18	16	4	128	15460	2
Modbus	0.5059	0.2278	0.5044	0.3138	0.82	1.68	4	16	1024	75	3
Motion_light	0.5023	0.1426	0.5030	0.2223	0.81	2.56	4	2	512	15961	2
Thermostat	0.5286	0.1289	0.4698	0.2023	0.71	2.41	4	5	1024	2508	3
Weather	0.5882	0.1914	0.6074	0.2911	1.15	3.39	8	24	2048	101	3
Global	0.6236	0.3345	0.3898	0.3601	1.64	2.39	4	16	2048	18259	2



ULEEN

- Binarização
 - Variável categórica de dois valores -> **Binarização Direta (0 e 1)**
 - Variável categórica complexa -> **One-Hot Encoding**
 - Variável numérica contínua -> **Termômetro Linear**
 - Variável numérica contínua de cauda longa -> **Termômetro Logarítmico**
 - Tratamento de NaN -> **Bit de Presença**
- Hiperparâmetros treinamento Globais:
 - Épocas: 30
 - Learning Rate: 0.0015
 - Batch Size: 64
 - Dropout: 0.0



ULEEN - Resultados

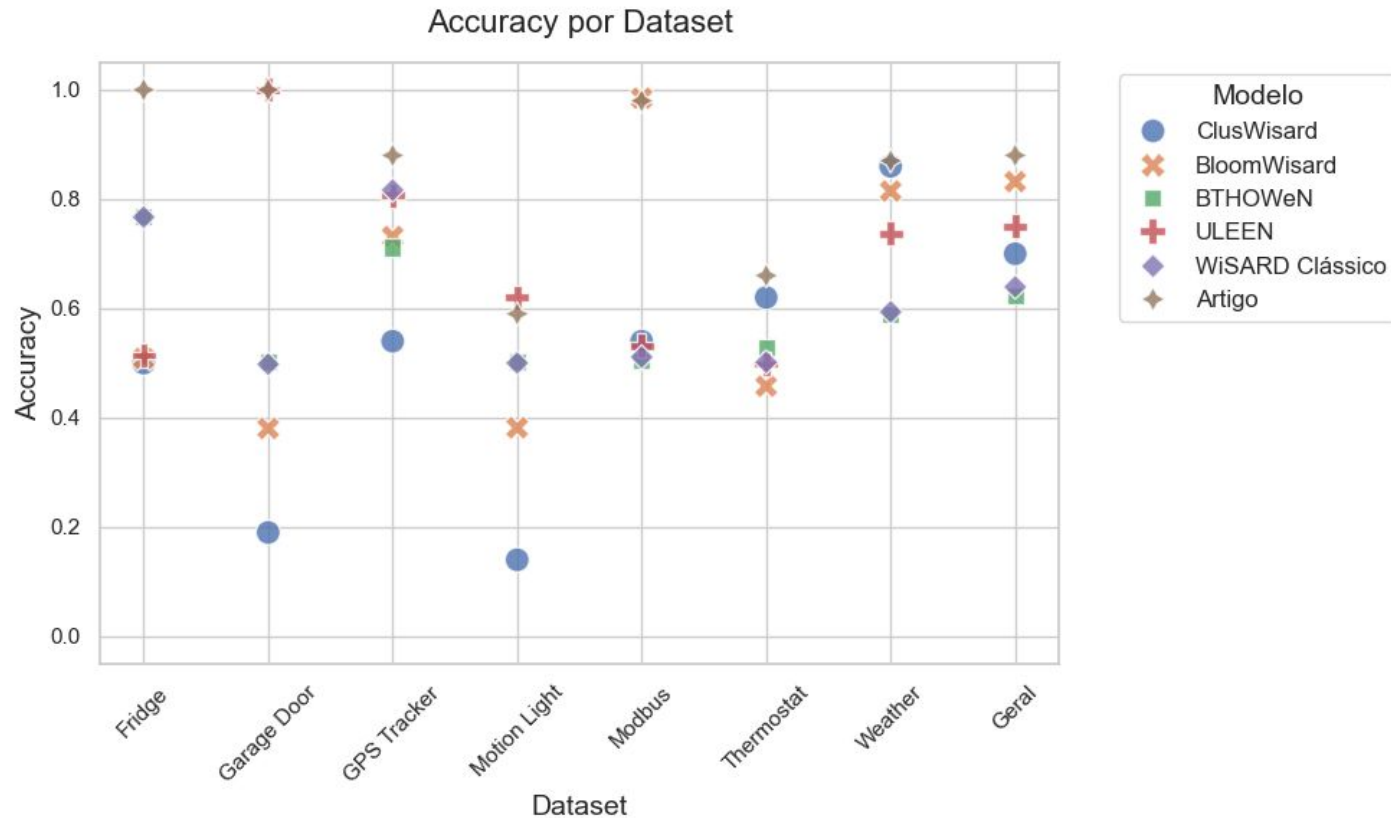
Dataset	Accuracy	Precision	Recall	F1-score	Train time (s)	Test time (s)	Bits/term	Bits/filtro	Entries/filtro
Fridge	0.5145	0.3739	0.5628	0.4484	89.43	1.10	8	7	128
Garage_door	1.0000	1.0000	1.0000	1.0000	85.26	0.10	24	4	32
Gps_tracker	0.8071	0.8711	0.8057	0.8371	96.41	0.10	24	14	128
Modbus	0.5318	0.5597	0.4831	0.5036	75.93	0.11	16	12	256
Motion_light	0.6201	0.6201	1.0000	0.7655	85.28	0.28	24	4	32
Thermostat	0.4984	0.4338	0.4733	0.3740	81.30	0.08	8	6	64
Weather	0.7374	0.4338	0.4733	0.3740	87.09	0.10	16	10	256
Global	0.7499	0.5373	0.6118	0.5691	874.22	1.66	24	18	256



ULEEN - Resultados

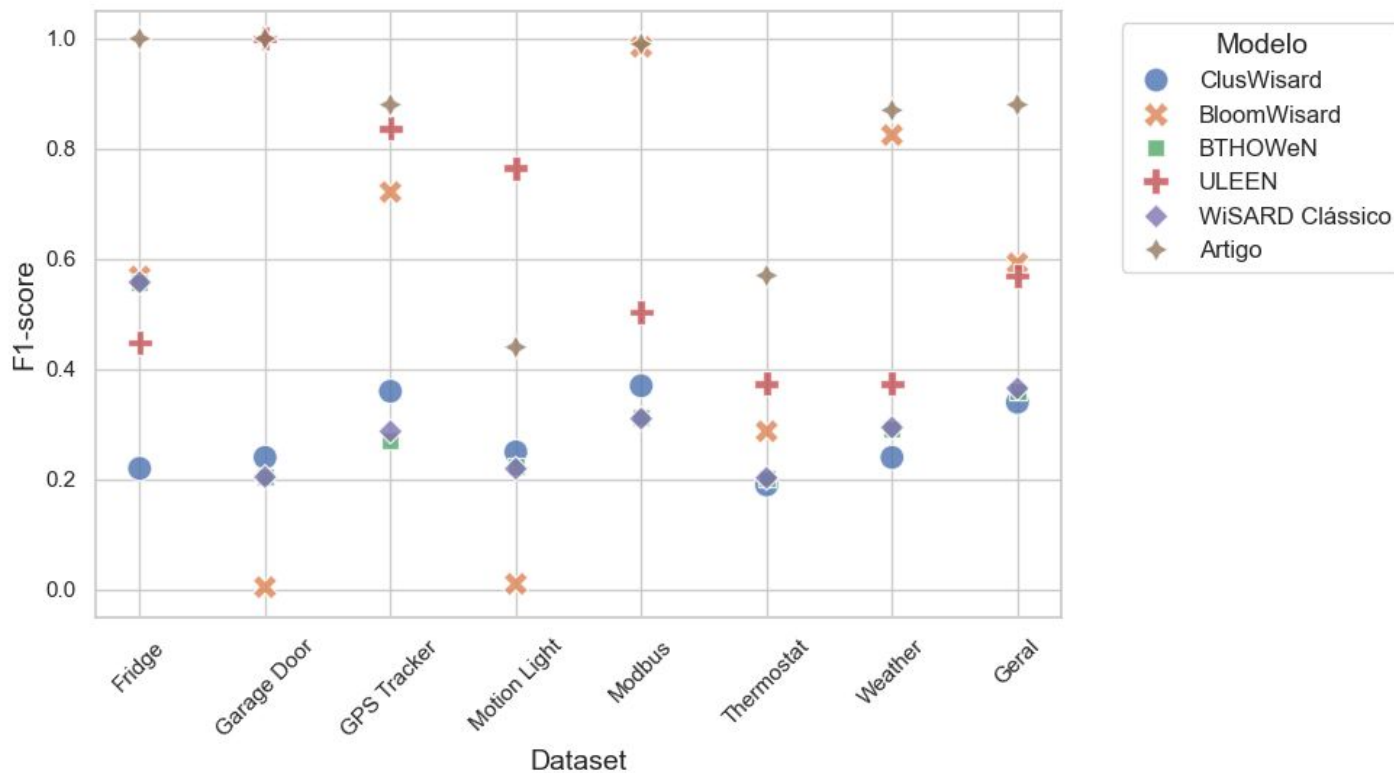
Dataset	Accuracy	Precision	Recall	F1-score	Train time (s)	Test time (s)	Bits/term	Bits/filtro	Entries/filtro
Fridge	0.5145 ± 0.1151	0.3739 ± 0.3053	0.5628 ± 0.4645	0.4484 ± 0.3668	89.43	1.10	8	7	128
Garage_door	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	85.26	0.10	24	4	32
Gps_tracker	0.8071 ± 0.0045	0.8711 ± 0.0043	0.8057 ± 0.0059	0.8371 ± 0.0040	96.41	0.10	24	14	128
Modbus	0.5318 ± 0.0159	0.5597 ± 0.0250	0.4831 ± 0.1487	0.5036 ± 0.0875	75.93	0.11	16	12	256
Motion_light	0.6201 ± 0.0000	0.6201 ± 0.0000	1.0000 ± 0.0000	0.7655 ± 0.0000	85.28	0.28	24	4	32
Thermostat	0.4984 ± 0.0377	0.4338 ± 0.2169	0.4733 ± 0.4472	0.3740 ± 0.3069	81.30	0.08	8	6	64
Weather	0.7374 ± 0.0088	0.4338 ± 0.2169	0.4733 ± 0.4472	0.3740 ± 0.3069	87.09	0.10	16	10	256
Global	0.7499 ± 0.0094	0.5373 ± 0.0192	0.6118 ± 0.0676	0.5691 ± 0.0218	874.22	1.66	24	18	256

Comparação dos Modelos - Acurácia

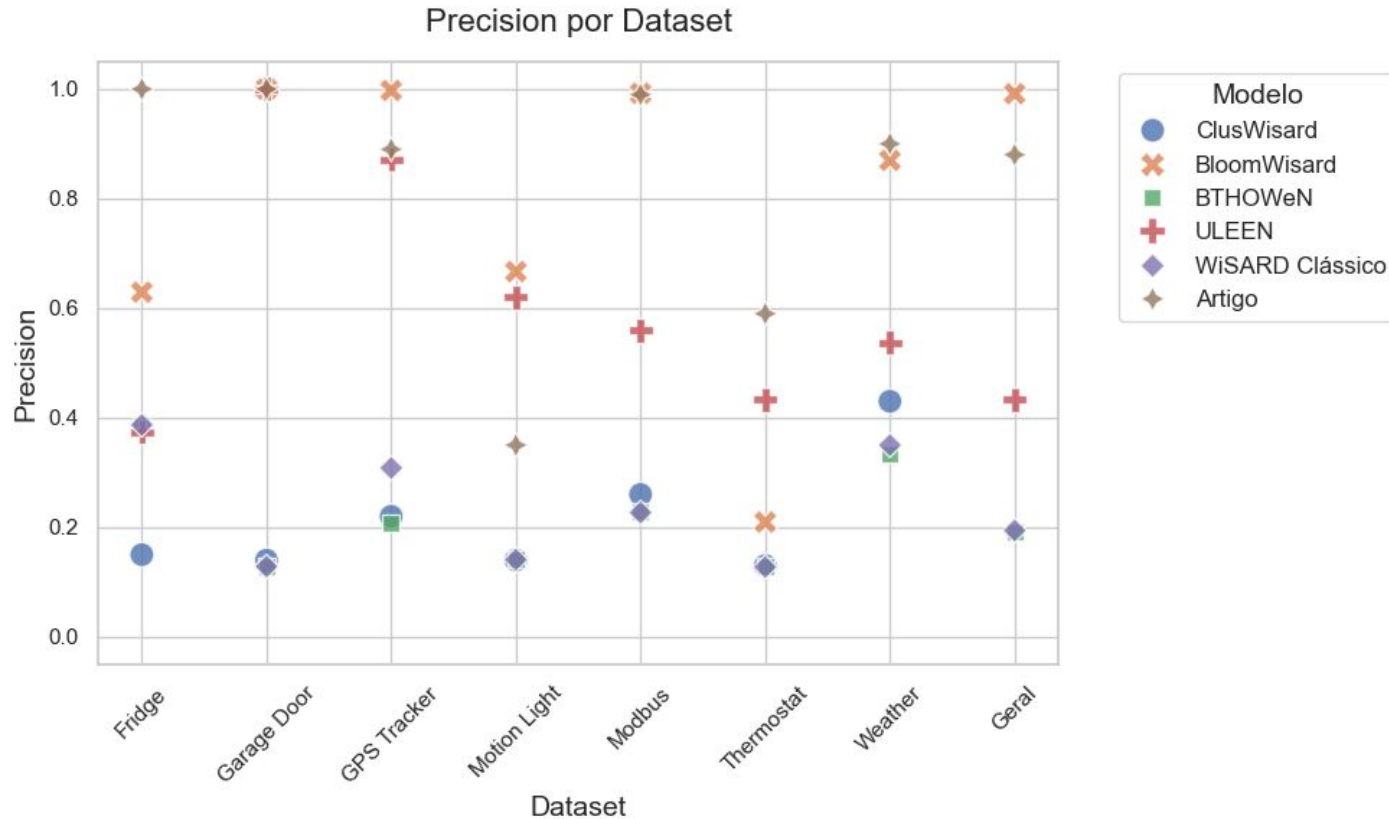


Comparação dos Modelos - F1-score

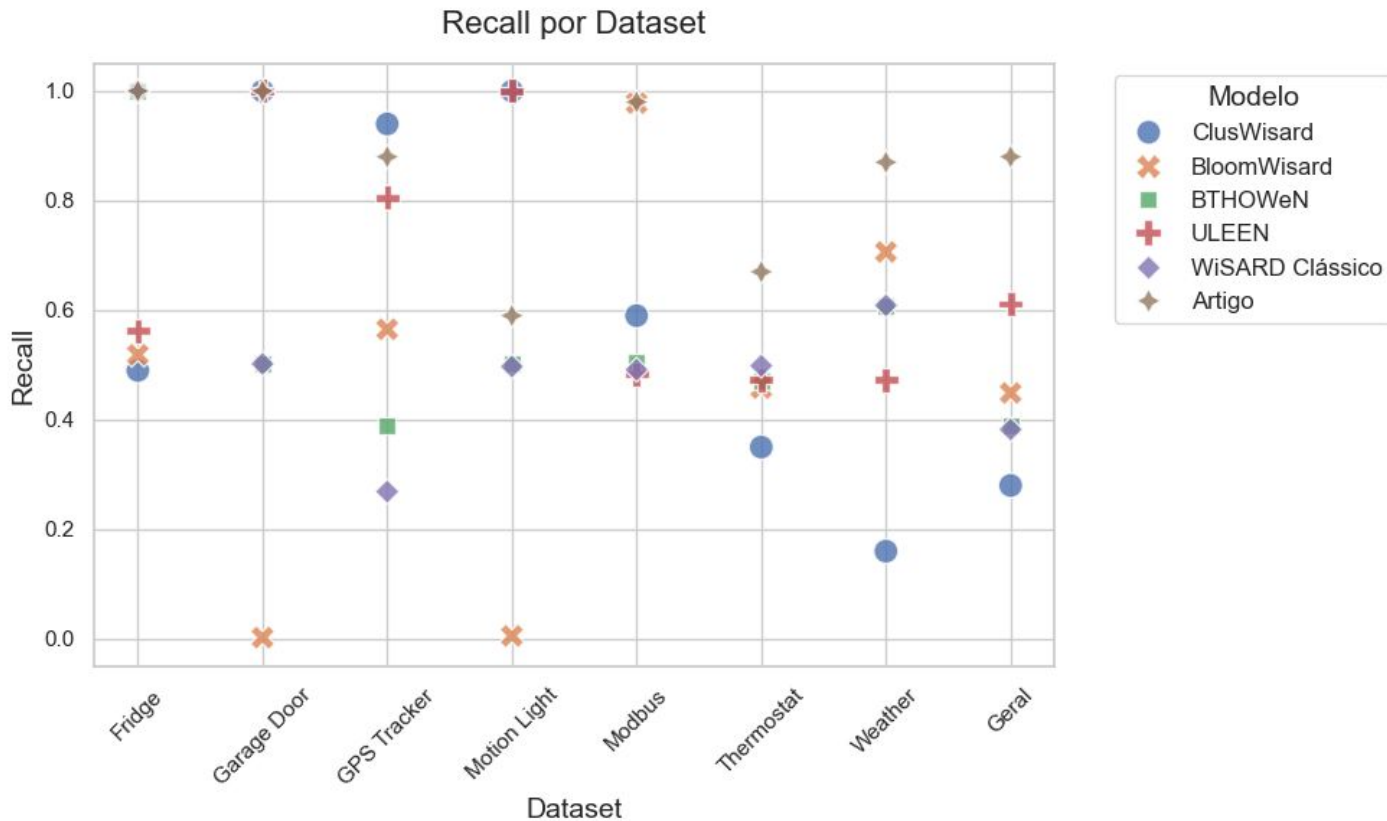
F1-score por Dataset



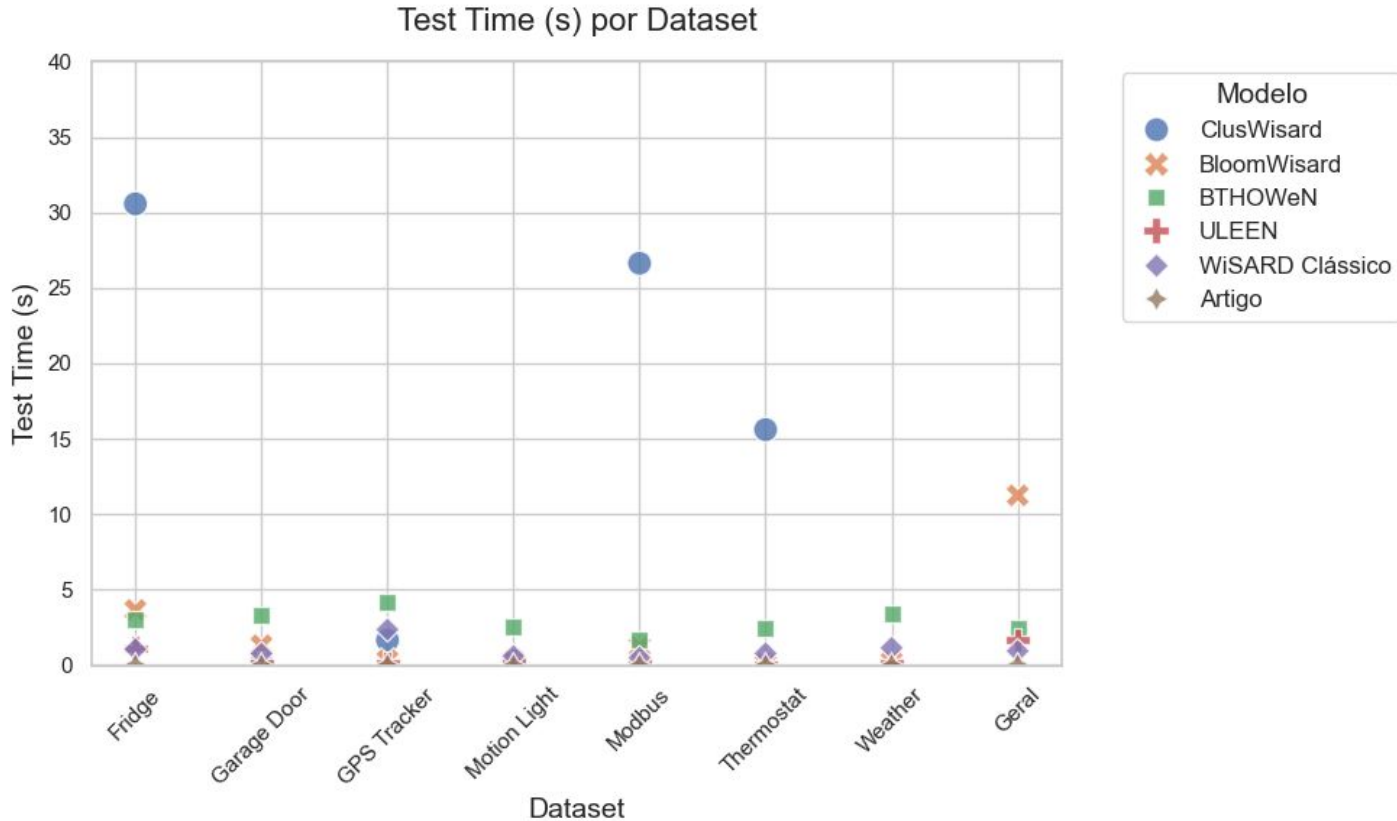
Comparação dos Modelos - Precisão



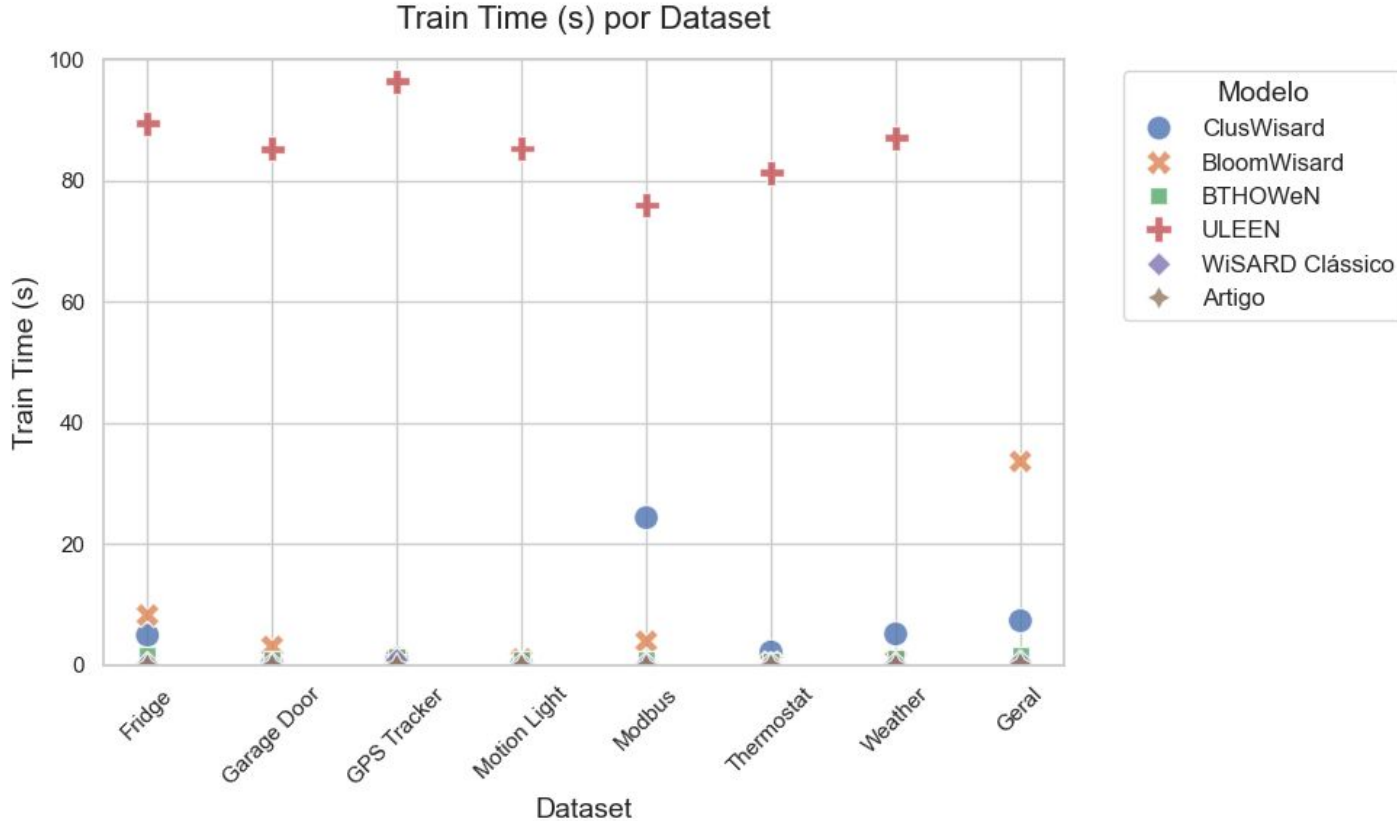
Comparação dos Modelos - Recall



Comparação dos Modelos - Test Time (s)



Comparação dos Modelos - Train Time (s)



Comparação dos Modelos - Fridge

Models	Accuracy	Precision	Recall	F-score	Train Time (s)	Test Time (s)
LR	0.57	0.34	0.58	0.43	0.017	0.001
LDA	0.77	0.79	0.77	0.77	0.004	0.001
kNN	0.99	0.99	0.99	0.99	0.147	2.556
RF	0.97	0.97	0.97	0.97	0.188	0.045
CART	0.97	0.97	0.97	0.97	0.032	0.005
NB	0.50	0.53	0.51	0.51	0.011	0.005
SVM	0.81	0.86	0.82	0.80	44.554	19.148
LSTM	1.00	1.00	1.00	1.00	190.493	3.705
ClusWisard	0.50	0.15	0.49	0.22	4.890	30.570
BloomWisard	0.51	0.63	0.52	0.57	8.220	3.640
BTHOWeN	0.77	0.39	1.00	0.56	1.470	3.010
ULEEN	0.51	0.37	0.56	0.45	89.430	1.100
WiSARD Clássico	0.77	0.39	1.00	0.56	0.178	1.038



Comparação dos Modelos - Garage Door

Models	Accuracy	Precision	Recall	F-score	Train Time (s)	Test Time (s)
LR	1.00	1.00	1.00	1.00	1.000	0.001
LDA	1.00	1.00	1.00	1.00	0.031	0.000
kNN	1.00	1.00	1.00	1.00	0.625	0.969
RF	1.00	1.00	1.00	1.00	0.062	0.000
CART	1.00	1.00	1.00	1.00	0.016	0.000
NB	1.00	1.00	1.00	1.00	0.010	0.002
SVM	1.00	1.00	1.00	1.00	30.558	6.180
LSTM	1.00	1.00	1.00	1.00	402.570	4.511
ClusWisard	0.19	0.14	1.00	0.24	0.120	0.120
BloomWisard	0.38	1.00	0.00	0.00	3.050	1.300
BTHOWeN	0.50	0.13	0.50	0.21	0.970	3.330
ULEEN	1.00	1.00	1.00	1.00	85.260	0.100
WISARD Clássico	0.50	0.13	0.50	0.20	0.075	0.743



Comparação dos Modelos - GPS Tracker

Models	Accuracy	Precision	Recall	F-score	Train Time (s)	Test Time (s)
LR	0.86	0.88	0.86	0.87	0.066	0.001
LDA	0.86	0.88	0.86	0.87	0.013	0.001
kNN	0.88	0.89	0.88	0.88	0.080	1.508
RF	0.85	0.85	0.85	0.85	0.833	0.099
CART	0.84	0.85	0.85	0.85	0.277	0.015
NB	0.84	0.86	0.85	0.86	0.009	0.007
SVM	0.86	0.88	0.87	0.87	25.127	9.483
LSTM	0.87	0.89	0.88	0.88	151.832	6.784
ClusWisard	0.54	0.22	0.94	0.36	0.990	1.640
BloomWisard	0.73	1.00	0.57	0.72	0.970	0.300
BTHOWeN	0.71	0.21	0.39	0.27	1.400	4.180
ULEEN	0.81	0.87	0.81	0.84	96.410	0.100
WiSARD Clássico	0.82	0.31	0.27	0.29	0.798	2.338



Comparação dos Modelos - Modbus

Models	Accuracy	Precision	Recall	F-score	Train Time (s)	Test Time (s)
LR	0.67	0.46	0.68	0.55	0.029	0.001
LDA	0.67	0.46	0.68	0.55	0.041	0.001
kNN	0.77	0.77	0.78	0.77	0.060	0.116
RF	0.97	0.98	0.98	0.98	1.587	0.031
CART	0.98	0.99	0.98	0.99	0.367	0.001
NB	0.67	0.46	0.68	0.55	0.012	0.002
SVM	0.67	0.46	0.68	0.55	19.286	3.558
LSTM	0.68	0.47	0.68	0.55	317.959	1.651
ClusWisard	0.54	0.26	0.59	0.37	24.320	26.630
BloomWisard	0.98	0.99	0.98	0.99	3.900	1.200
BTHOWeN	0.51	0.23	0.50	0.31	0.820	1.680
ULEEN	0.53	0.56	0.48	0.50	75.930	0.110
WiSARD Clássico	0.51	0.23	0.49	0.31	0.100	0.470



Comparação dos Modelos - Motion Light

Models	Accuracy	Precision	Recall	F-score	Train Time (s)	Test Time (s)
LR	0.58	0.34	0.59	0.43	0.028	0.001
LDA	0.58	0.34	0.59	0.43	0.036	0.001
kNN	0.54	0.34	0.59	0.43	3.157	6.409
RF	0.58	0.34	0.59	0.43	0.100	0.008
CART	0.58	0.34	0.59	0.43	0.008	0.001
NB	0.58	0.34	0.59	0.43	0.011	0.002
SVM	0.58	0.34	0.59	0.43	30.686	6.259
LSTM	0.59	0.35	0.59	0.44	63.132	3.730
ClusWisard	0.14	0.14	1.00	0.25	0.080	0.080
BloomWisard	0.38	0.67	0.01	0.01	0.970	0.300
BTHOWeN	0.50	0.14	0.50	0.22	0.810	2.560
ULEEN	0.62	0.62	1.00	0.77	85.280	0.280
WiSARD Clássico	0.50	0.14	0.50	0.22	0.066	0.577

Comparação dos Modelos - Thermostat

Models	Accuracy	Precision	Recall	F-score	Train Time (s)	Test Time (s)
LR	0.66	0.44	0.66	0.53	0.030	0.001
LDA	0.66	0.44	0.66	0.53	0.035	0.001
kNN	0.60	0.56	0.61	0.57	0.064	0.088
RF	0.66	0.59	0.66	0.53	1.044	0.023
CART	0.59	0.56	0.59	0.57	0.009	0.001
NB	0.66	0.44	0.66	0.53	0.009	0.002
SVM	0.66	0.44	0.66	0.53	69.340	10.246
LSTM	0.66	0.45	0.67	0.54	69.073	1.820
ClusWisard	0.62	0.13	0.35	0.19	2.140	15.600
BloomWisard	0.46	0.21	0.46	0.29	0.240	0.070
BTHOWeN	0.53	0.13	0.47	0.20	0.710	2.410
ULEEN	0.50	0.43	0.47	0.37	81.300	0.080
WiSARD Clássico	0.50	0.13	0.50	0.20	0.108	0.760



Comparação dos Modelos - Weather

Models	Accuracy	Precision	Recall	F-score	Train Time (s)	Test Time (s)
LR	0.58	0.60	0.59	0.53	0.083	0.001
LDA	0.60	0.59	0.60	0.53	0.041	0.000
kNN	0.81	0.81	0.81	0.81	0.066	0.414
RF	0.84	0.84	0.84	0.84	0.789	0.008
CART	0.87	0.88	0.87	0.87	0.258	0.030
NB	0.69	0.72	0.69	0.67	0.011	0.002
SVM	0.63	0.68	0.63	0.55	65.469	8.029
LSTM	0.82	0.82	0.81	0.80	318.242	3.790
ClusWisard	0.86	0.50	0.16	0.24	5.110	71.290
BloomWisard	0.81	0.99	0.71	0.82	0.770	0.230
BTHOWeN	0.59	0.19	0.61	0.29	1.150	3.390
ULEEN	0.74	0.43	0.47	0.37	87.090	0.100
WiSARD Clássico	0.59	0.19	0.61	0.29	0.175	1.126



Comparação dos Modelos - Geral

Models	Accuracy	Precision	Recall	F-score	Train Time (s)	Test Time (s)
LR	0.61	0.37	0.61	0.46	3.023	0.005
LDA	0.68	0.74	0.68	0.62	1.041	0.004
kNN	0.84	0.85	0.84	0.84	58.018	109.361
RF	0.85	0.87	0.85	0.85	10.884	0.164
CART	0.88	0.90	0.88	0.88	6.308	0.022
NB	0.62	0.63	0.62	0.51	0.210	0.069
SVM	0.61	0.37	0.61	0.46	3525.052	558.663
LSTM	0.81	0.83	0.81	0.80	1596.809	9.023
ClusWisard	0.70	0.43	0.28	0.34	7.300	817.100
BloomWisard	0.83	0.87	0.45	0.59	33.590	11.240
BTHOWeN	0.62	0.33	0.39	0.36	1.640	2.390
ULEEN	0.75	0.54	0.61	0.57	874.220	1.660
WiSARD Clássico	0.64	0.35	0.38	0.37	0.470	0.920



Conclusão

- O modelo ULEEN demonstrou bons resultados comparativos com o artigo original, alcançando 100% de acurácia em cenários específicos.
- As WNNs apresentaram tempos de treinamento consideravelmente menores que modelos de Deep Learning (LSTM), viabilizando o uso em hardware IoT restrito.
- Foi verificado que a estratégia de binarização apresentou impacto significativo no desempenho do modelo WNN treinado.
- Embora modelos como KNN liderem em acurácia pura, as WNNs oferecem o melhor balanço entre performance e economia de recursos.